

Final Report

20210638 최무송

Overview

- UN Regulation 155
- Wifi protocol
- Remote Keyless Entry
- Phone Projection
- VehicleSec'25 paper review

UN Regulation 155

- UN Regulation 155
 - Cyber Security as Approval
- CMCS(Cyber Security **Management** System)
- Annex 5
 - Threats/Attacks and Vulnerabilities - Mitigations
 - What to focus?

Wifi protocol

- 802.11: Technical Standard for Wireless LAN
- EAP/EAPoL
 - 4-way handshake (MSK $\rightarrow$ PMK $\rightarrow$ PTK)
 - $PTK = PRF(PMK + ANonce + SNonce + MAC(AA) + MAC(SA))$
 - PMKID Dictionary Attack on M1
 - KRACK (Key Reinstallation Attack on M3)

Wifi protocol

- Encryption
 - WEP(RC4, static key)
 - WPA(TKIP)
 - WPA2(AES-CCMP)
 - WPA3(Dragonfly(hash-to-curve))
- More Attacks
 - Fragattacks
 - Dragonblood(timing side-channel offline dictionary/DoS)

Remote Keyless Entry

- RKE: Vehicle – digital key transmission
- NFC, BLE, UWB
- RRSI(Received signal strength indication) vs. Time-of-Flight
- Attacks
 - Replay Attack, Rolljam Attack, Rollback Attack
 - Relay Attack
 - Distance Reduction Attack

Phone Projection

- Phone projection to IVI(In-Vehicle Infotainment)
- MirrorLink
 - Heap overflow → CAN bus access
- Apple Carplay
 - USB connection (recognition) → USB role switch → iAP2(Inter-Accessory Protocol 2) session → CarPlay Session
 - iPhone centered Authentication
- Android Auto
 - Possible Vulnerabilities

Paper Review: A Black Box System for Automotive Digital Forensics

- Current Black box
 - trigger-based recording
 - inadequate time synchronization
 - insufficient trust preservation
- Suggestion
 - GPS-based time synchronization
 - continuous Electronic Control Unit authentication
- Example - sudden brake failure
 - driver error (through input logs)
 - manufacturing defects (through ECU performance history)
 - malicious interference (through ECU authentication logs)